

YAN SHUO TAN

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ACADEMIC APPOINTMENTS	Assistant Professor	2022—
	Department of Statistics and Data Science, National University of Singapore	
	• Faculty Affiliate, Institute of Data Science • Faculty Affiliate, NUS AI Institute	
	Neyman Visiting Assistant Professor	2020—2021
	Department of Statistics, University of California, Berkeley	
	• <i>Mentor: Prof. Bin Yu</i>	
	Post-doctoral Fellow	2021—2022
	Department of Statistics, University of California, Berkeley	
	• <i>Mentor: Prof. Bin Yu</i>	
	Post-doctoral Fellow	2018—2019
	Department of Statistics, University of California, Berkeley	
	• <i>Mentor: Prof. Bin Yu</i>	
	Patrick J. McGovern Research Fellow	2018—2018
	Simons Institute, University of California, Berkeley	
EDUCATION	Ph.D., Mathematics	2013—2018
	University of Michigan, Ann Arbor, USA	
	• Advisors: <i>Prof. Roman Vershynin & Prof. Anna Gilbert</i>	
	B.S., Mathematics (Honors)	2009—2013
	University of Chicago, Chicago, USA	
RESEARCH INTERESTS	Theoretical and applied aspects of statistical machine learning and data science.	
	• Theory and methodology for tree-based methods and ensembles • Bayesian tree ensemble modeling • In-context learning for tabular data	
GRANTS	MOE AcRF Tier 1 Grant (NUS Cross Faculty Grant) A-8004458-00-00, \$250,000	2026
	MOE AcRF Tier 1 Grant A-8002498-00-00, \$250,000	2024
	NUS Start-up Grant A-8000448-00-00, \$180,000	2022
ACHIEVEMENTS & AWARDS	NUS Inauguration Grant	2022
	US Junior Oberwolfach Fellowship	2019
	Patrick J. McGovern Research Fellowship	2018

Paul R. Cohen Prize

2013

Awarded to graduating seniors with the highest record in mathematics (5 awarded per year)

Phi Beta Kappa

2012

JOURNAL
PUBLICATIONS

(* equal contribution or alphabetical ordering, † student/postdoc supervised)

1. **Yan Shuo Tan**, Jason M Klusowski, and Krishnakumar Balasubramanian, “Statistical-Computational Trade-offs for Recursive Adaptive Partitioning Estimators”, *Annals of Statistics*, 2026.
2. Qiong Zhang^{*†}, **Yan Shuo Tan**^{*}, and Jiahua Chen, “Byzantine-Tolerant Distributed Learning of Finite Mixture Models”, *Journal of the Royal Statistical Society, Series B*, 2026.
3. **Yan Shuo Tan**, Chandan Singh, Keyan Nasser, Abhineet Agarwal, James Duncan, Omer Ronen, Matthew Epland, Aaron Kornblith, and Bin Yu, “Fast Interpretable Greedy-Tree Sums”, *Proceedings of the National Academy of Sciences*, vol. 122, no. 7, pp. e2310151122, 2025.
4. **Yan Shuo Tan** and Roman Vershynin, “Online Stochastic Gradient Descent with Arbitrary Initialization Solves Non-Smooth, Non-Convex Phase Retrieval”, *Journal of Machine Learning Research*, vol. 24, no. 58, pp. 1–47, 2023.
5. John Lipor, David Hong, **Yan Shuo Tan**, and Laura Balzano, “Erratum to: Subspace Clustering Using Ensembles of K-Subspaces”, *Information and Inference: A Journal of the IMA*, vol. 12, no. 1, pp. 590–590, 2023.
6. Chandan Singh, Keyan Nasser, **Yan Shuo Tan**, Tiffany Tang, and Bin Yu, “imodels: A Python Package for Fitting Interpretable Models”, *Journal of Open Source Software*, vol. 6, no. 61, pp. 3192, 2021.
7. John Lipor, David Hong, **Yan Shuo Tan**, and Laura Balzano, “Subspace Clustering Using Ensembles of K-Subspaces”, *Information and Inference: A Journal of the IMA*, vol. 10, no. 1, pp. 73–107, 2021.
8. Nick Altieri^{*}, Rebecca L Barter^{*}, James Duncan^{*}, Raaz Dwivedi^{*}, Karl Kumbier^{*}, Xiao Li^{*}, Robert Netzorg^{*}, Briton Park^{*}, Chandan Singh^{*}, **Yan Shuo Tan**^{*}, Tiffany Tang^{*}, Yu Wang^{*}, Chao Zhang^{*}, and Bin Yu^{*}, “Curating a COVID-19 Data Repository and Forecasting County-Level Death Counts in the United States”, *Harvard Data Science Review*, 2020.
9. Raaz Dwivedi^{*}, **Yan Shuo Tan**^{*}, Briton Park, Mian Wei, Kevin Horgan, David Madigan, and Bin Yu, “Stable Discovery of Interpretable Subgroups via Calibration in Causal Studies”, *International Statistical Review*, vol. 88, pp. S135–S178, 2020.
10. **Yan Shuo Tan** and Roman Vershynin, “Phase Retrieval via Randomized Kaczmarz: Theoretical Guarantees”, *Information and Inference: A Journal of the IMA*, vol. 8, no. 1, pp. 97–123, 2019.
11. **Yan Shuo Tan**, “Energy Optimization for Distributions on the Sphere and Improvement to the Welch Bounds”, *Electronic Communications in Probability*, 2017.

CONFERENCE
PUBLICATIONS

12. Ruizhe Deng[†], Bibhas Chakraborty^{*}, Ran Chen^{*}, and **Yan Shuo Tan**^{*‡}, “BFTS: Thompson Sampling with Bayesian Additive Regression Trees”, *International Conference on Machine Learning*, 2026 [Spotlight].
13. Jean Feng, Avni Kothari, Luke Zier, Chandan Singh, and **Yan Shuo Tan**, “Bayesian Concept Bottleneck Models with LLM Priors”, *Advances in Neural Information Processing Systems*, 2025.

14. Abhineet Agarwal*, **Yan Shuo Tan***, Omer Ronen, Chandan Singh, and Bin Yu, “Hierarchical Shrinkage: Improving the Accuracy and Interpretability of Tree-Based Models”, *International Conference on Machine Learning*, pp. 111–135, 2022 [**Long presentation**].
15. **Yan Shuo Tan**, Abhineet Agarwal, and Bin Yu, “A Cautionary Tale on Fitting Decision Trees to Data from Additive Models: Generalization Lower Bounds”, *International Conference on Artificial Intelligence and Statistics*, pp. 9663–9685, 2022.
16. **Yan Shuo Tan** and Roman Vershynin, “Polynomial Time and Sample Complexity for Non-Gaussian Component Analysis: Spectral Methods”, *Conference on Learning Theory*, pp. 498–534, 2018.

PREPRINTS

17. Zineng Xu[†], Subhroshekhar Ghosh*, and **Yan Shuo Tan***, “On the Statistical Optimality of Optimal Decision Trees”, submitted to *Annals of Statistics*.
18. Tianqi Zhao[†], Guanyang Wang, **Yan Shuo Tan**, and Qiong Zhang[†], “TabClustPFN: A Prior-Fitted Network for Tabular Data Clustering”, submitted to *ICML*.
19. Jean Feng, Avni Kothari, Patrick Vossler, Andrew Bishara, Lucas Zier, Newton Addo, Aaron Kornblith, **Yan Shuo Tan**, and Chandan Singh, “Human-AI Co-design for Clinical Prediction Models”, major revision at *Nature Digital Medicine*.
20. Ruinan Jin, Gexin Huang, Xinwei Shen, Qiong Zhang[†], **Yan Shuo Tan**, and Xiaoxiao Li, “See-in-Pairs: Reference Image-Guided Comparative Vision-Language Models for Medical Diagnosis”, submitted to *Medical Image Analysis*.
21. Qiong Zhang^{*†}, **Yan Shuo Tan***, Qinglong Tian*, and Pengfei Li, “TabPFN: One Model to Rule Them All?”, major revision at *Journal of the American Statistical Association*.
22. Xin Chen^{*†}, Jason M Klusowski*, **Yan Shuo Tan***, and Chang Yu*, “Revisiting Randomization in Greedy Model Search”, submitted to *ICML*.
23. **Yan Shuo Tan**, Omer Ronen, Theo Saarinen, and Bin Yu, “The Computational Curse of Big Data for Bayesian Additive Regression Trees: A Hitting Time Analysis”, in revision at *Annals of Statistics*.
24. Abhineet Agarwal*, Ana M Kenney*, **Yan Shuo Tan***, Tiffany M Tang*, and Bin Yu, “Integrating Random Forests and Generalized Linear Models for Improved Accuracy and Interpretability”, in revision at *Journal of Machine Learning Research*.
25. Xin Chen^{*†}, Jason M Klusowski*, and **Yan Shuo Tan***, “Error Reduction from Stacked Regressions”, submitted to *Journal of the American Statistical Association*.
26. Omer Ronen*, Theo Saarinen*, **Yan Shuo Tan***, James Duncan, and Bin Yu, “A Mixing Time Lower Bound for a Simplified Version of BART”, arXiv preprint, 2022.
27. **Yan Shuo Tan**, “Sparse Phase Retrieval via Sparse PCA Despite Model Misspecification: A Simplified and Extended Analysis”, arXiv preprint, 2017.

IN
PREPARATION

28. Zineng Xu[†], Yuchao Cai[†], and **Yan Shuo Tan[†]**, “On Early Stopping and Ideal Spatial Adaptation for CART”.
29. Chandan Singh, **Yan Shuo Tan**, Weijia Xu, Zelalem Gero, Weiwei Yang, Michel Galley, and Jianfeng Gao, “AGENTIC-IMODELS: Evolving Interpretability Tools for Agents”.
30. Ruizhe Deng[†], Sumetha Loganathan[†], Kenyon Ng, Bibhas Chakraborty, Qiong Zhang[†], and **Yan Shuo Tan[†]**, “Solving Contextual Bandits via Efficient Online Uncertainty Quantification with Tabular Foundation Models”.
31. Tianqi Zhao[†], Qiong Zhang^{*†}, and **Yan Shuo Tan***, “Uncertainty Evaluation and Calibration for Tabular Foundation Models”.

INVITED TALKS

- (1) Understanding the Statistical Gaps in Tree-Based Learning

 - Gouxionghui Seminar, Online
 - MBZUAI Seminar, Abu Dhabi, UAE
 - UniMelb Seminar, Melbourne, Australia

2025–2026
- (2) Sparse Modeling Revisited: From Linear Models to LLMs

 - AIVP, RIKEN-AIP & A*STAR Joint Workshop, Tokyo, Japan
 - ICSDS, Seville, Spain

2025
- (3) TabPFN: One Model to Rule Them All?

 - EcoSta, Tokyo, Japan

2025
- (4) Lessons on Mixing for Bayesian Additive Regression Trees

 - BayesComp, Singapore
 - Workshop on Efficient Sampling Algorithms for Complex Models, IMS, National University of Singapore, Singapore

2025
- (5) Statistical-Computational Trade-offs for Recursive Adaptive Partitioning Estimators

 - One World MINDS Seminar, Online
 - Renmin University of China International Forum on Statistics, Beijing, China
 - INFORMS International Meeting, Singapore
 - RIKEN-AIP Deep Learning Theory Team Seminar, Tokyo, Japan

2024–2025
- (6) A Mixing Time Lower Bound for a Simplified Version of BART

 - Workshop on Optimization in the Big Data Era, IMS, National University of Singapore, Singapore
 - BIRS Workshop on Harnessing the Power of Latent Structure Models and Modern Big Data Learning, Hangzhou, China
 - IMS Asia-Pacific Rim Meeting, Melbourne, Australia
 - Workshop on the Mathematics of Data, IMS, National University of Singapore, Singapore

2022–2024
- (7) MDI+: A Flexible Random Forest-Based Feature Importance Framework

 - Young Mathematical Scientists Forum, IMS, National University of Singapore, Singapore

2023
- (8) Understanding and Overcoming the Statistical Limitations of Decision Trees

 - Neyman Seminar, Department of Statistics, University of California, Berkeley, USA
 - Department Seminar, Department of Statistics and Data Science, National University of Singapore, Singapore
 - Workshop on Machine Learning and its Applications, IMS, National University of Singapore, Singapore
 - Workshop on Information Theory and Data Science, IMS, National University of Singapore, Singapore
 - ESD Research Seminar, Singapore University of Technology and Design, Singapore

2022–2023
- (9) Stable Discovery of Interpretable Subgroups via Calibration in Causal Studies

 - Bernoulli-IMS Young Researchers Pre-meeting, Seoul, South Korea
 - ICSA Canada Chapter Symposium, Banff, Canada
 - Causal Inference Research Group Seminar, University of California, Berkeley, USA
 - Biostatistics Seminar, University of California, Berkeley, USA

2020–2022
- (10) Implicit Regularization for Constant Step-Size SGD: Why It Works for Non-Smooth, Non-Convex, Low Rank Models

2019

- Foundations of Data Science Program Reunion Workshop, Simons Institute, Berkeley, USA
- AMS Spring Central and Western Joint Sectional Meeting, Special Session on Sparsity, Randomness, and Optimization, Honolulu, USA
- ACO Seminar, University of California, Irvine, USA

(11) Efficient Algorithms for Phase Retrieval in High Dimensions 2017–2018

- BLISS Seminar, University of California, Berkeley, USA
- Math-FLDS Seminar, University of Southern California, USA
- Probability Seminar, University of California, Irvine, USA
- ARC-TRIAD Seminar, Georgia Institute of Technology, USA
- SILO Seminar, University of Wisconsin, USA

TEACHING
EXPERIENCE

National University of Singapore 2023–

- **ST5209/X**: Analysis of Time Series Data (Semester II 2023, Semester II 2024, Semester II 2025, Semester II 2026)
- **ST5201X**: Statistical Foundations of Data Science (Semester I 2022)

University of California, Berkeley 2021–

- **Data 102**: Data, Inference, and Decisions (Spring 2021)
 - Capstone course for the data science major. Develops the probabilistic foundations of inference in data science and builds a comprehensive view of the modeling and decision-making life cycle. Topics include: frequentist and Bayesian decision-making, false discovery rate, causal inference, general linear modeling, non-parametric modeling, bandits, and reinforcement learning.
- **Stat 210B**: Theoretical Statistics II (Spring 2020)
 - Second course on theoretical statistics for PhD students, focusing on high-dimensional statistics.

University of Michigan semester)–

- Precalculus (Fall 2013 – Winter 2018 (1 semester))
 - Section of 20–25 students; responsible for lecturing, group work, materials, and grading.
- Calculus I (Fall 2013 – Winter 2018 (3 semesters))
 - Section of 20–25 students; responsible for lecturing, group work, materials, and grading.
- Calculus II (Fall 2013 – Winter 2018 (2 semesters))
 - Section of 20–25 students; responsible for lecturing, group work, materials, and grading.
- Multivariate Calculus (Fall 2013 – Winter 2018 (1 semester))
 - Led lab tutorial sessions and graded.
- Differential Equations (Fall 2013 – Winter 2018 (1 semester))
 - Led lab tutorial sessions and graded.
- Precalculus (Course Co-coordinator) (Fall 2013 – Winter 2018 (1 semester))
 - Co-coordinated a course of ~500 students. Trained and supervised new instructors; wrote homework and exams.

Students Advised

• Duan Shuo (BS)	2026–2028 (<i>expected</i>)
• Zhao Yaya (PhD)	2025–2026 (<i>expected</i>)
• Le Thi Hong Ha (PhD)	2025–2029 (<i>expected</i>)
• Martin Eppert (PhD), <i>co-advised with Subhroshekhar Ghosh</i>	2025–2029 (<i>expected</i>)
• Sumetha Loganathan (PhD), <i>co-advised with Bibhas Chakraborty</i>	2025–2027 (<i>expected</i>)
• Cai Yuchao (Postdoc)	2024–2026 (<i>expected</i>)
• Deng Ruizhe (PhD), <i>co-advised with Bibhas Chakraborty</i>	2023–2027 (<i>expected</i>)
• Zhuang Tianyi (PhD)	2023–2027 (<i>expected</i>)
• Xu Zineng (PhD)	2023–2026 (<i>expected</i>)
• Zeng Zitong (MSc)	2023–2024
• Wang Yihe (BS)	2025–2025
• Song Yida (BS)	2023–2024

Thesis Advisory Committee

• Wenshan Qu (National University of Singapore)	2024–2027 (<i>expected</i>)
• Nicolas Alexander Ihlo (University of Regensburg)	2025–2027 (<i>expected</i>)

Other Mentoring

• BAIR Undergraduate Mentoring Program, University of California, Berkeley	2020–2020
• Mentoring junior PhD students in Yu Group, University of California, Berkeley	2019–2022

Reviewing Activities

• Annals of Applied Statistics	2020
• Annals of Statistics	2025
• Biometrika	2024
• Conference on Learning Theory	2018, 2019
• Conference on Neural Information Processing Systems	2023, 2025
• Foundations of Computer Science	2018
• IEEE Transactions on Information Theory	2023, 2026
• IMA Journal of Numerical Analysis	2017
• Information and Inference: A Journal of the IMA	2019
• International Conference on Artificial Intelligence and Statistics	2021, 2024
• Journal of Data Science	2024
• Journal of Machine Learning Research	2025
• Journal of the American Statistical Association	2024, 2025
• Journal of the Royal Statistical Society: Series B	2022, 2024
• Proceedings of the National Academy of Sciences	2020
• Scandinavian Journal of Statistics	2026
• Science China	2025
• SMAI Journal of Computational Mathematics	2017
• Statistical Methods in Medical Research	2025
• Symposium on Discrete Algorithms	2019, 2020

Other Service

• Graduate Research Committee Member, Department of Statistics and Data Science, National University of Singapore	2024–2026
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- Co-organizer, FIM-IMS Joint Workshop on Mathematics for Data Science, National University of Singapore *2026–2026*
- Co-organizer, IMS New Researchers Conference Asia, Institute of Mathematical Statistics *2026–2026*
- Young Researcher Committee Member, Bernoulli Society *2024–2026*
- Group Meeting Organizer, Yu Group, University of California, Berkeley *2020–2020*
- Student Probability Seminar Organizer, University of Michigan *2016–2017*
- Student Geometry Seminar Organizer, University of Michigan *2014–2015*